

Deep Learning Model based Human Activity Recognition using Kinect Sensor Data

1. Introduction

Human Activity Recognition (HAR) has become an important research area in computer vision, human–computer interaction, and ubiquitous computing, driven by the need to understand human behavior in applications such as healthcare monitoring, rehabilitation, intelligent surveillance, and assistive living. Over the past decade, the Microsoft Kinect sensor has played a central role in advancing HAR research because of its ability to capture synchronized RGB images, depth maps, and 3D skeletal joint positions in real time. Unlike wearable sensors, which require physical contact with the user, or RGB-only vision systems that are sensitive to lighting and background variations, the Kinect provides a rich multimodal representation of human motion while maintaining simplicity, low cost, and non-intrusiveness. These characteristics have made Kinect-based HAR a widely adopted platform for studying both controlled laboratory activities and natural, unconstrained human behavior.

The introduction of Kinect’s real-time skeletal tracking marked a major shift in HAR methodology. Instead of relying solely on color or depth images, researchers could now directly access a structured, articulated representation of the human body through tracked joint coordinates. This allowed activity recognition models to focus on pose transitions, joint trajectories, and temporal motion patterns without the need for complex background subtraction or dense optical flow computations. Early studies demonstrated that even simple handcrafted features derived from these joint coordinates—such as joint displacement, inter-joint distances, or summary statistics—were highly effective for distinguishing between daily activities and gestures. These results showed that the geometry of skeletal motion itself carries strong discriminative information, establishing Kinect skeletons as a robust modality for activity analysis.

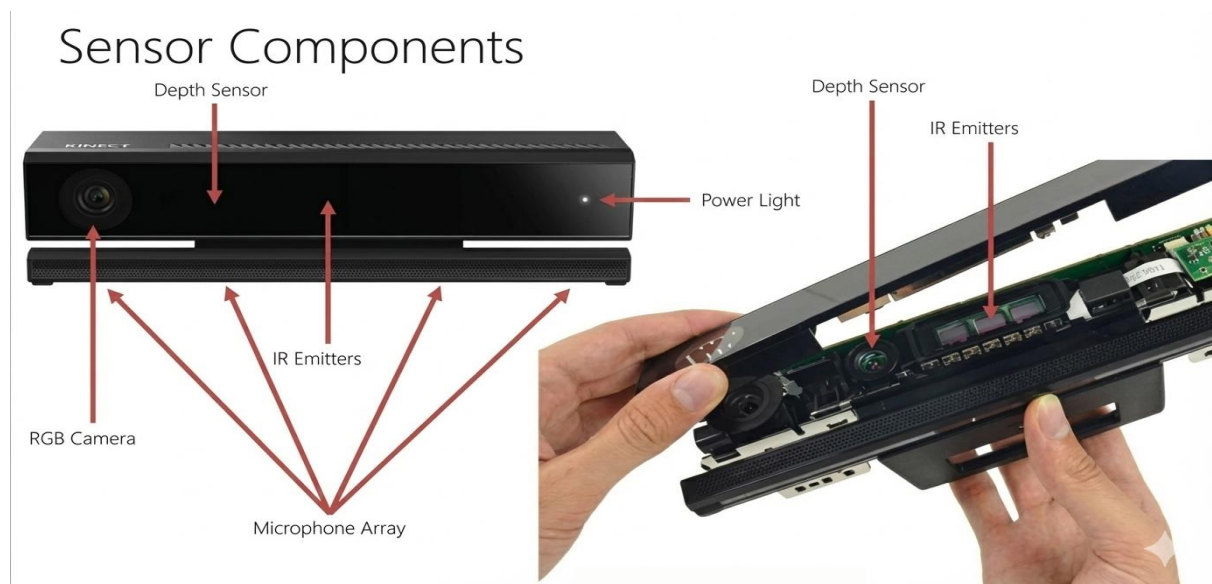


Figure 1. Microsoft Kinect Sensor Xbox One

As recognition tasks became more complex, researchers began exploring depth-map descriptors, temporal modeling frameworks, and hybrid learning pipelines. Depth sequences enabled the extraction of geometric cues that complemented skeletal trajectories, improving recognition performance in scenarios involving subtle limb movements or object interactions. Researchers also experimented with Hidden Markov Models, Support Vector Machines, and clustering-based pipelines to capture posture dynamics and temporal transitions. These classical approaches established the foundations of Kinect-based HAR, but they relied heavily on manual feature design and often struggled with multi-person interactions, occlusions, or long-term temporal dependencies.

With the rise of deep learning, Kinect-based HAR progressed into a new phase in which models could automatically learn both spatial and temporal representations. Convolutional neural networks were used to extract hierarchical patterns from depth images and spatial skeleton encodings, while recurrent neural networks—particularly LSTMs and BiLSTMs—became essential for modeling long-range temporal structure in activity sequences. Hybrid CNN–RNN architectures, attention mechanisms, and multimodal fusion strategies further improved performance by capturing motion patterns at multiple scales and by combining complementary information sources such as RGB, depth, skeletal joints, and object context. These developments resulted in state-of-the-art performance on many Kinect-based benchmarks and established deep learning as the dominant methodology for HAR.

In parallel, the broader HAR community explored concepts such as multi-scale temporal processing, residual skip pathways, and gated recurrent mechanisms. These architectures were designed to address common challenges in sequential modeling, such as vanishing gradients, loss of long-term contextual information, and the need to capture both fine-grained and coarse-grained motion patterns. These ideas have strongly influenced recent Kinect-based HAR models, particularly those built on multi-scale recurrent structures and gated residual pathways, which aim to enhance temporal feature flow and stabilize deep sequence learning.

Although alternative device-free HAR technologies such as Wi-Fi sensing and radar sensing have gained attention for their privacy-preserving characteristics, Kinect-based HAR remains one of the most widely studied approaches due to its balance of accuracy, interpretability, and multimodal sensing ability. The combination of depth imaging and articulated skeleton tracking continues to provide a rich platform for activity modeling, especially in applications requiring detailed understanding of human posture and movement.

2. Related Work

Human Activity Recognition (HAR) has emerged as a prominent research domain, attracting considerable attention from the computer vision and machine learning communities. Over the past decade, numerous studies have proposed diverse methodologies—ranging from classical machine learning approaches to sophisticated deep learning architectures—for recognizing and

classifying human activities from various sensor modalities, including RGB cameras, depth sensors, inertial measurement units (IMUs), and wearable devices. This section provides a comprehensive review of the existing literature that is most relevant to the present study.

2.1 Survey and Review Studies on HAR

Kaseris et al. [1] presented a comprehensive survey on deep learning methods applied to human activity recognition. The authors concluded that while vision-based HAR systems can effectively address environmental challenges such as varying lighting conditions, future performance improvements will depend significantly on high-quality data normalization techniques and the continued paradigm shift toward automatic feature extraction through deep neural networks. The survey further highlighted that vision sensors, such as Microsoft Kinect, provide skeletal data representations in lower-dimensional spaces, thereby facilitating swifter processing and enabling real-time applications in domains such as surveillance and eldercare.

Similarly, Hossen and Abas [17] conducted an extensive review of machine learning techniques for HAR, examining state-of-the-art methods and emerging trends. They concluded that future research must prioritize scalability and adaptability to real-world environments in order to overcome persistent challenges such as sensor noise and class imbalance. Their review highlighted that advanced frameworks like PoseConv3D, which utilizes three-dimensional heatmap volumes, achieved strong average accuracies across benchmark datasets, including 97.1% on the NTU60 dataset. The authors emphasized the continued need for innovation and interdisciplinary collaboration to advance HAR technology.

Sarveshwaran et al. [14] investigated the application of deep learning for human activity recognition and concluded that, although deep learning significantly advances HAR capabilities, unresolved problems remain regarding real-world interference and noise. Their analysis demonstrated that ensemble learning models reduce pre-processing time by supporting automatic feature extraction. Specifically, they reported that ResNet v2 achieved high accuracy scores of 91% and 92% across the UCI 101 and HMDB 51 video datasets, respectively.

2.2 Hybrid Deep Learning Architectures for HAR

A substantial body of research has focused on developing hybrid deep learning models that combine the strengths of multiple architectures to improve activity recognition performance. Lalwani and Ganeshan [2] proposed a novel CNN-BiLSTM-GRU hybrid deep learning model for human activity recognition, which achieved a remarkable accuracy of 99.7% on the Wireless Sensor Data Mining (WISDM) dataset. The authors concluded that parameter optimization is critical for model performance, specifically identifying a batch size of 0.3 as the optimal setting for improving model robustness and convergence. The results, validated

through ANOVA statistical testing, confirmed that the model is statistically reliable and highly effective for real-world applications in bipedal robotics and healthcare.

Challa et al. [3] introduced a multibranch CNN-BiLSTM model for human activity recognition using wearable sensor data. The authors concluded that their proposed multibranch architecture, which employs convolutional kernels of different sizes across parallel branches, is highly effective at capturing various temporal local dependencies inherent in sensor signals. The model achieved impressive classification accuracies of 96.05% on the WISDM dataset, 96.37% on the UCI-HAR dataset, and 94.29% on the PAMAP2 dataset. Notably, they highlighted that the tri-branch configuration maintains a superior trade-off between recognition accuracy and computational cost compared to single-branch or quad-branch model variants.

Tokas et al. [5] reported on a real-time deployable attention-driven CNN-LSTM framework for human activity recognition using wearable sensors. Their system achieved an average accuracy of 99.5% across three public benchmarks, including a peak accuracy of 99.7% on the mHealth dataset. The authors concluded that the integration of a Convolutional Block Attention Module (CBAM) and self-attention mechanism allows the model to refine spatial and temporal features by focusing selectively on the most informative frames. Furthermore, the application of structured pruning and quantization techniques successfully produced a lightweight model suitable for real-time deployment on smartphones without degrading recognition performance.

Khatun et al. [8] proposed a deep CNN-LSTM architecture augmented with a self-attention mechanism for human activity recognition using wearable sensor data. The authors concluded that their hybrid model efficiently extracts discriminative features from raw sensor signals, achieving a 99.93% accuracy on their personalized H-Activity dataset. The model also demonstrated robustness on established benchmark datasets, reaching 98.76% on MHEALTH and 93.11% on UCI-HAR. They demonstrated that the self-attention mechanism significantly enhances the model's predictive capabilities and could be effectively applied in clinical settings for monitoring elderly health.

Garcia-Gonzalez et al. [36] investigated deep learning models for real-life human activity recognition from smartphone sensor data and concluded that deep learning algorithms represent the optimal choice for HAR in real-life environments, surpassing traditional machine learning results with a peak accuracy of 94.80%. They found that a 90-second sliding window size and the combination of accelerometer and gyroscope sensors provided the highest-quality results. Their DS-CNN-LSTM hybrid model proved particularly effective, especially when individual sensor streams were evaluated independently before concatenation.

Sain et al. [20] proposed a hybrid CNN-BiLSTM deep learning model for human action recognition in indoor environments. The authors concluded that their architecture achieves state-of-the-art results, reaching 99.5% accuracy on the LNMIIT-KHAD dataset. They highlighted that the model's reliance on skeletal joint angles and kinematic features makes it exceptionally robust against variations in illumination and cluttered backgrounds. Furthermore, the model demonstrated strong generalization capability across diverse sensing conditions and extended accurately to activities outside the training set.

2.3 Graph-Based and Attention-Based Approaches

Graph-based methods have gained significant traction in recent years due to their ability to naturally represent the structural topology of the human body. Dai et al. [6] investigated the application of Graph Convolutional Networks (GCNs) for human action recognition using skeletal features. The authors concluded that GCNs are inherently more expressive than traditional vector-based methods because they naturally model the human body's graph-like structure, where joints serve as vertices and bones serve as edges. The research indicated that GCN-based models, such as the Spatial-Temporal Graph Convolutional Network (ST-GCN), achieve significant performance enhancements on large-scale datasets, reaching accuracies up to 96.2% on the NTU RGB+D dataset. The authors further suggested that GCNs provide a solid foundation for future multimodal fusion strategies in action recognition.

Gosala et al. [9] proposed GCN-LSTM, a hybrid graph convolutional network model originally developed for schizophrenia classification from EEG signals, which demonstrated the powerful capability of combining graph-based spatial modeling with temporal sequence learning. The model achieved an average accuracy of 99.25% across 8-second epoch data segments. The authors concluded that the hybrid GCN-LSTM architecture successfully captures both the spatial relationships between EEG channels and the temporal evolution of signals simultaneously. The study noted that the model outperformed all compared state-of-the-art methods and suggested its potential as a valuable clinical decision-support tool.

2.4 Skeleton-Based and Kinect-Based Recognition Systems

The use of skeletal data captured from depth sensors, particularly Microsoft Kinect, has been extensively explored for human activity and gait recognition. Saini et al. [4] introduced a novel coarse-to-fine framework for continuous human activity recognition using Kinect, which segments continuous activity sequences into sitting and standing groups before performing fine-level recognition within each group. The system achieved a continuous sequence recognition rate of 68.9% and a coarse-level grouping accuracy of 93.81% using a Bidirectional LSTM neural network (BLSTM-NN). The study demonstrated that modeling temporal transitions between activities is essential for real-life applications and that their segmented approach significantly outperforms traditional isolated activity recognition methods.

Açış and Güney [7] investigated the classification of human movements using the Kinect sensor and concluded that joint (skeletal) data provides substantially more successful results than image data for movement classification tasks. They found that using an LSTM-based feature extractor combined with a Support Vector Machine (SVM) classifier achieved a maximum success rate of 98.85%, which was vastly superior to classical machine learning applied to raw data (72.60%) and transfer learning approaches (61.58%). The study further emphasized that elbow joints play a particularly significant role in determining movement direction and were identified as the most critical joints for accurate classification.

Reddy and Chattopadhyay [10] proposed a human activity recognition system using Kinect-captured data with a stick model representation. The authors concluded that their selective four-joint stick model, focusing on the head and shoulders, effectively reduces system complexity and noise introduced by unreliable joint estimations. The results demonstrated that combining range and mean coordinate features outperformed individual feature sets, achieving a maximum accuracy of 97.29% on their custom dataset and 94.06% on the MSR 3D Daily Activity dataset. The authors expressed their intention to integrate this system for monitoring elderly individuals in home settings.

Kumar et al. [11] addressed human activity recognition in real-time environments using skeleton joints and concluded that using only 10–12% of gesture frames is sufficient for identifying activities with excellent accuracy in real-time scenarios. Their system, utilizing Principal Component Analysis (PCA) for dimensionality reduction and Artificial Neural Networks (ANN) for classification, achieved overall accuracies ranging from 94% to 98%. The authors emphasized that speed, accuracy, and robustness are often contradictory requirements, and their approach provides a high-performance balance particularly suited for single-object recognition tasks.

Komang et al. [13] explored human activity recognition using skeleton data and support vector machines. The authors concluded that the distance of the subject from the sensor significantly impacts recognition accuracy. Their SVM-based system achieved its highest average accuracy of 93.75% when the subject was positioned at 2 meters from the Kinect sensor. The study introduced a multi-class SVM with a radial basis function (RBF) kernel, which provided accurate results for eight specific indoor and standing activities.

Ashwini et al. [19] developed a skeletal data-based activity recognition system and concluded that SVM is a more effective classifier than K-Nearest Neighbors (KNN) for three-dimensional skeletal data on the KARD dataset. The SVM classifier achieved an average overall accuracy of 98%, which was 3.5% higher than KNN. They reported that sensitivity reached 100% for

certain actions, demonstrating that dimensionally reduced features obtained via PCA can still provide highly accurate activity identification.

Yadav et al. [35] investigated skeleton-based human activity recognition using ConvLSTM and guided feature learning. The authors concluded that the sequential fusion of CNN and LSTM networks is superior to isolated or parallel model configurations, as it captures the optimal set of spatiotemporal features from skeleton-based data. Their ConvLSTM model achieved 98.89% accuracy, which represented a 5–6% improvement over individual CNN (93.89%) or LSTM (92.75%) networks. The derived "guided features" effectively preserved user privacy while remaining independent of camera orientation or clothing variations.

2.5 Gait Recognition and Modeling

Gait analysis constitutes another closely related area of research that has been studied extensively in the context of human motion understanding. Bari and Gavrilova [15] proposed KinectGaitNet, a Kinect-based gait recognition system using deep convolutional neural networks. The authors concluded that their system can accurately identify individuals using residual learning-based CNNs without requiring resampling of the gait cycle. The model achieved a rank-1 recognition accuracy of 99.33% on the KGB dataset and 96.91% on the UPCV dataset. They determined that Global Average Pooling (GAP) is the most effective pooling method for handling variable-dimensional feature maps in gait recognition tasks.

Abhayasinghe and Murray [12] investigated human gait modeling, prediction, and classification for level walking using harmonic models derived from a single thigh-mounted IMU. The authors concluded that human gait phases can be accurately modeled and classified using harmonic models extracted from a single inertial sensor. They observed that normalized overtone amplitudes are generally smaller for males than females in their sample population, although phase characteristics remain comparable across genders. The research provided a successful framework for predicting gait patterns in natural, unconstrained environments.

Qin et al. [18] investigated human gait prediction and recognition algorithms for lower limb-assisted exoskeleton robots. The researchers concluded that combining fuzzy inference with least-squares support vector regression (LS-SVR) significantly reduces system delay in gait phase detection. The results demonstrated high accuracy in gait data prediction, reaching 99.93% for specific pressure sensor configurations. The proposed algorithm effectively recognizes four specific gait phases—heel landing, full foot landing, heel off-ground, and toe off-ground—enabling smoother and more natural exoskeleton assistance.

2.6 Multi-Scale and Residual Learning Approaches

Guo et al. [16] proposed MRC-LSTM, a hybrid approach combining multi-scale residual CNN and LSTM networks for Bitcoin price prediction. Although the application domain differs from HAR, the underlying architectural principle of utilizing multi-scale residual blocks to extract rich features across multiple temporal scales is directly relevant to activity recognition. The authors concluded that the multi-scale residual block effectively strengthens the representational ability of the feature extraction pipeline. The MRC-LSTM model achieved the lowest prediction error, with a mean absolute error (MAE) of 166.52 compared to isolated CNN, LSTM, or MLP models. The empirical results further confirmed the model's strong generalization ability to predict short-term trends for other time-series data.

2.7 Summary and Research Gaps

The reviewed literature demonstrates substantial progress in human activity recognition across multiple modalities and architectures. However, several research gaps remain. First, many existing skeleton-based models rely on conventional feature engineering or isolated deep learning architectures that do not fully exploit both the spatial topology and temporal dynamics of skeletal sequences simultaneously. Second, while GCN-based methods effectively capture structural relationships, they are often computationally expensive and may not integrate seamlessly with temporal sequence models. Third, most existing hybrid models have not incorporated multi-scale residual learning with graph-based spatial modeling and bidirectional temporal encoding in a unified framework. The present study addresses these gaps by proposing an integrated architecture that synergistically combines multi-scale residual feature extraction, graph convolutional spatial modeling, and bidirectional LSTM-based temporal encoding for robust and accurate human activity recognition from Kinect skeletal data.

3. Methodology

3.1 Dataset Description

3.1.1 Skeletal Dataset Description

For this study, a **custom skeletal dataset** was created using the Microsoft Kinect v2 sensor to capture human motion for Human Activity Recognition (HAR). The dataset captures high-fidelity skeletal motion, providing 3D coordinates of major body joints over time, which are critical for accurate modeling of human activities. Data were collected from multiple participants performing a diverse set of everyday activities, enabling detailed analysis of both posture and motion dynamics.

A. Key characteristics of the dataset:

- **Sensor Technology:** Microsoft Kinect v2, enabling real-time skeletal tracking.
- **Data Representation:** 3D coordinates (x, y, z) for 25 major joints.
- **Participants:** 30 volunteers representing diverse body types and ages.
- **Activity Scope:** Seven activities encompassing both dynamic and static motions.

- Temporal Coverage: Sufficient duration to capture meaningful movement patterns.
- Frame Rate: Approximately 30 Hz, providing high temporal resolution for precise motion analysis.

Table 1 Dataset Analysis

Feature / Metric	Custom Skeletal Dataset	KARD Dataset
Sensor	Kinect v2	Kinect v2
Number of Subjects	30	10
Number of Activities	7	18
Number of Samples	129,981	954,450
Sampling Rate (Hz)	30.07	30
Avg Sequence Length (Frames)	618.96	1767.5
Number of Skeletal Joints	25	15
Joint Representation	3D Cartesian (x,y,z)	3D Cartesian (x,y,z)
Skeleton Format	Kinect v2	Kinect v2
Train/Val/Test Split (%)	70/15/15	70/15/15
Evaluation Protocol	Subject-Independent Split	Subject-Independent Split
Target Application	Human Activity Recognition for Robot Imitation	Human Activity Recognition
Benchmark Usage	Yes	Yes

B. Skeletal Joint Representation from Kinect

The Kinect sensor records the spatial positions of 25 key joints, forming a complete skeletal representation for each participant. Joint positions are expressed in meters relative to the sensor, allowing precise measurement of inter-joint relationships and fine-grained motion. Such comprehensive skeletal data are essential for distinguishing between complex or similar activities.

Tracked joints include:

SpineBase, SpineMid, Neck, Head, ShoulderLeft, ElbowLeft, WristLeft, HandLeft,

ShoulderRight, ElbowRight, WristRight, HandRight, HipLeft, KneeLeft, AnkleLeft, FootLeft, HipRight, KneeRight, AnkleRight, FootRight, SpineShoulder, HandTipLeft, ThumbLeft, HandTipRight, ThumbRight.

Additional details:

- Each frame includes timestamps and body ID, ensuring proper association with individual participants.
- Data capture encompasses both subtle and pronounced movements, enabling extraction of features such as joint trajectories, angular velocities, and motion dynamics.
- The dataset is structured to support activity segmentation and temporal modelling, facilitating downstream HAR tasks.

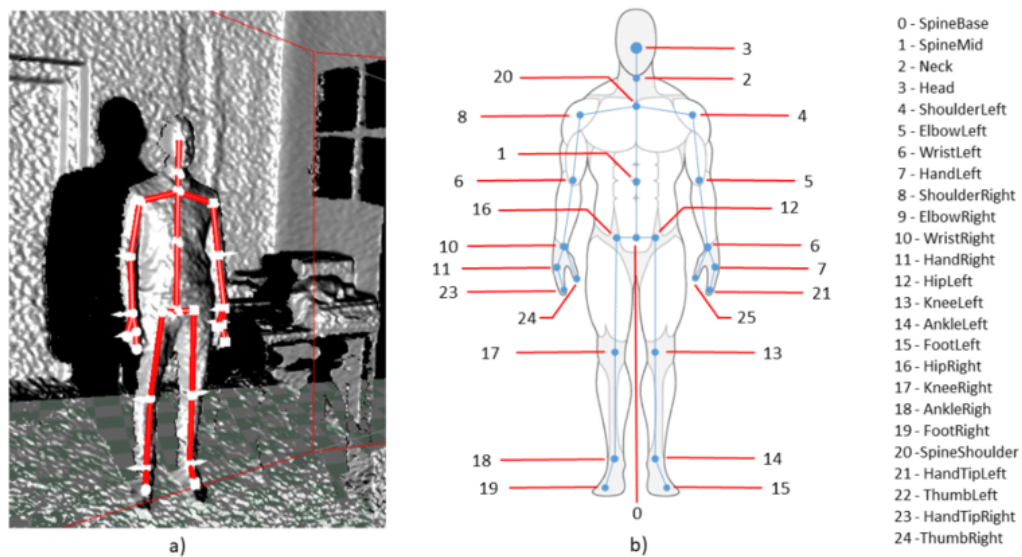


Figure 2. Skeletal Joint Representation

C. Dataset Structure and Activity Classes

The dataset is organized hierarchically, with a dedicated folder for each participant. Within each folder, separate CSV files correspond to the seven recorded activities: bending, jumping, running, sitting, squat, standing, and walking. Each file contains time-stamped 3D coordinates for all 25 joints, providing a complete representation of the participant's motion for the respective activity.

Participant diversity in terms of body structure, height, weight, and age ensures that the dataset captures natural variations in movement patterns. This heterogeneity enhances the generalizability of models trained on the dataset, reflecting realistic population variability. The structured format simplifies access, processing, and analysis of skeletal data, supporting both spatial and temporal modeling of human motion.

3.1.2 KARD Dataset Description

The **Kinect Activity Recognition Dataset (KARD)** is a widely used benchmark for Human Activity Recognition, providing multi-modal recordings of human motion captured with the Microsoft Kinect sensor. The dataset includes 18 distinct activities performed three times by ten different subjects, offering variability in execution style and providing a comprehensive resource for evaluating HAR algorithms.

A. Activity Classes

KARD contains 18 actions representing a diverse set of dynamic and daily-life activities: horizontal arm wave, high arm wave, two-hand wave, catch cap, high throw, draw X, draw tick, toss paper, forward kick, side kick, take umbrella, bend, hand clap, walk, phone call, drink, sit down, stand up. Each activity is repeated three times by ten subjects, resulting in a total of 2160 files.

B. Data Organization

The dataset follows a standardized naming convention to uniquely identify each recording, in the form `aA_sS_eN_string`, where A is the two-digit action ID, S is the two-digit subject ID, N is the repetition number, and string indicates the data modality. The dataset provides four types of files for each sequence: `depthmaps.txt` for depth map sequences, `.mp4` for RGB video at 640×480 resolution, `realworld.txt` for skeletal joint positions in real-world coordinates, and `screen.txt` for skeletal joint positions in screen coordinates with depth information. For example, `a04_s03_e02_realworld.txt` contains the real-world joint coordinates for the second repetition of action #4 performed by subject #3.

C. Skeletal Data Representation

The dataset records 15 key skeletal joints in each frame, listed consecutively as: head, neck, right shoulder, right elbow, right hand, left shoulder, left elbow, left hand, torso, right hip, right knee, right foot, left hip, left knee, left foot. Each file contains $15 \times F$ lines, where F is the number of frames in the sequence, and each line reports three numbers: (x, y, z) for real-world coordinates in `realworld.txt` or (u, v, depth) for screen coordinates in `screen.txt`. This structure captures both fine-grained and larger body movements for precise motion analysis.

D. Temporal and Spatial Properties

KARD consists of 540 sequences, totaling approximately one hour of video captured at 30 frames per second with 640×480 pixel resolution. The dataset provides sufficient temporal coverage and spatial resolution to enable accurate modeling of human activities, supporting both temporal and spatial analysis in Human Activity Recognition systems.

Table 2 : Activity Distribution Comparison

Activity	Custom Dataset (Num Samples)	Custom Dataset (%)	KARD Dataset (Num Samples)	KARD Dataset (%)
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Standing / Stand up	19,174	14.74%	41,490	4.35%
Bending / Bend	19,012	14.63%	46,365	4.86%
Jumping	18,780	14.45%	-	-
Sitting / Sit down	18,365	14.13%	50,385	5.28%
Squat	18,312	14.10%	-	-
Walking / Walk	18,279	14.06%	96,270	10.09%
Running	18,059	13.89%	-	-
Catch Cap	-	-	48,210	5.05%
Draw Tick	-	-	51,885	5.44%
Draw X	-	-	58,605	6.14%
High Throw	-	-	40,170	4.21%
Drink	-	-	55,935	5.86%
Forward Kick	-	-	39,780	4.17%
Hand Clap	-	-	39,555	4.14%
High Arm Wave	-	-	50,925	5.33%
Horizontal Arm Wave	-	-	50,700	5.31%
Phone Call	-	-	78,285	8.20%
Take Umbrella	-	-	61,530	6.45%
Toss Paper	-	-	43,995	4.61%
Two Hand Wave	-	-	61,440	6.44%
Total	129,981	100%	954,450	100%

3.2 Data Preprocessing

The data preprocessing pipeline converts raw Kinect skeletal recordings into fixed-length, standardized sequences suitable for the MS-GRS-BiLSTM model. The process begins with loading and cleaning each CSV file by removing non-essential fields and reconstructing missing joint values. Recordings are then segmented into overlapping 64-frame windows using a stride of 16, with zero-padding applied when needed to ensure uniform length. All features are normalized using Z-score standardization to maintain consistent scaling across joints. Optional Gaussian noise is added to improve robustness against sensor variations. Activity labels are encoded into integer and one-hot formats, and the final dataset is split using either window-level sampling or person-level partitioning to support reliable performance evaluation.

Table 3 : Overall Skeletal Joint Statistics Comparison

Dataset	Axis	Min (avg across joints)	Max (avg across joints)	Mean (avg across joints)	Std (avg across joints)
Custom Dataset	X	-1.758	0.766	0.007	0.250
	Y	-0.962	1.274	-0.006	0.253
	Z	1.191	4.633	2.611	0.441
KARD Dataset	X	-1068.204	1504.217	76.95	332.91
	Y	-1330.955	1166.143	46.57	176.64
	Z	1738.799	3489.051	2804.55	144.87

3.2.1 Data Loading and Cleaning

A. Numeric Feature Extraction (CSV to Numeric Matrix)

Each Kinect activity recording is first loaded from its corresponding CSV file, where every row represents a single skeleton frame captured over time. The data is imported into a structured dataframe, after which the timestamp and body ID fields are removed because they do not contribute to the motion dynamics being modeled. The remaining numerical columns—representing the 3D joint coordinates—form the core feature set, resulting in an F -dimensional vector (for example, $F = 75$ when 25 joints are tracked across the x, y, and z axes).

B. Missing Value Handling (Data Cleaning)

Since uninterrupted skeletal data is essential for stable learning, any missing or corrupted joint values are corrected through a sequential filling strategy. Forward-fill and backward-fill operations reconstruct incomplete frame sequences, and any unresolved gaps are replaced with zeros to ensure consistency across all recordings.

C. Output Representation

After cleaning, each recording is transformed into a well-structured numerical matrix of shape

$$\text{arr} \in \mathbb{R}^{T_{\text{rec}} \times F},$$

where T_{rec} denotes the total number of frames in the sequence. Each row in this matrix represents a complete and fully cleaned skeletal posture, ready for the subsequent windowing and normalization stages.

3.2.2 Sequence Windowing and Padding

A. Temporal Segmentation Using Sliding Windows

To prepare the Kinect skeletal recordings for sequence-based learning, each cleaned data stream is segmented into fixed-length windows using a sliding-window approach. A window length of $T = 64$ frames and a stride of $s = 16$ frames are employed, resulting in roughly 75% overlap between adjacent windows. This deliberate overlap helps maintain temporal continuity and significantly increases the number of training samples available to the MS-GRS-BiLSTM model.

B. Window Extraction Procedure

Each skeletal recording is represented as

$$\mathbf{X}_{\text{numeric}} \in \mathbb{R}^{T_{\text{rec}} \times F},$$

where T_{rec} is the number of frames in the recording and F denotes the number of joint-coordinate features. Windows are generated according to

$$\mathbf{X}_{\text{window}}^{(i)} = \mathbf{X}_{\text{numeric}}[i \cdot s : i \cdot s + T, :],$$

with i indicating the window index. The total number of windows derived from a single file is computed as

$$N_w^{(\text{file})} = \lfloor \frac{T_{\text{rec}} - T}{s} \rfloor + 1.$$

This formulation ensures consistent extraction of fixed-length temporal segments whenever $T_{\text{rec}} \geq T$.

C. Padding for Short Sequences

In cases where a recording is shorter than the required window length ($T_{\text{rec}} < T$), zero-padding is applied to extend the sequence to exactly T frames. This guarantees that even brief activity clips contribute valid windows, preventing the loss of information from shorter motion episodes.

D. Output Representation

After applying windowing and padding across all recordings, the resulting segments are aggregated into a unified 3D tensor:

$$\mathbf{X} \in \mathbb{R}^{N_w \times T \times F},$$

where N_w is the total number of extracted windows. This tensor provides a standardized and temporally consistent representation that serves as the direct input to subsequent normalization and training stages of the MS-GRS-BiLSTM model.

3.2.3 Feature Normalization (Z-Score Standardization)

A. Purpose of Normalization

Feature normalization is applied to ensure that all skeletal coordinates contribute uniformly during model training. In this work, Z-score standardization is implemented using the *StandardScaler*, which transforms each feature to have zero mean (μ) and unit variance (σ). This prevents the MS-GRS-BiLSTM model from overemphasizing coordinates with naturally larger numeric ranges, thereby improving numerical stability, convergence speed, and overall model robustness.

B. Input Representation and Flattening

The normalization stage operates on the fully segmented window dataset represented as:

$$\mathbf{X} \in \mathbb{R}^{N_w \times T \times F},$$

where N_w is the number of windows, T is the fixed temporal length, and F is the feature dimension.

To compute global statistics, the temporal axis is temporarily flattened:

$$\mathbf{X}_{\text{flat}} \in \mathbb{R}^{(N_w \cdot T) \times F}.$$

Flattening ensures that each feature's distribution is evaluated across all time steps and windows consistently.

C. Computation of Mean and Standard Deviation

Using *StandardScaler*, the following statistics are computed for each feature $f \in \{1, \dots, F\}$:

Mean:

$$\mu_f = \frac{1}{N_w T} \sum_{i=1}^{N_w T} x_{i,f}$$

Standard Deviation:

$$\sigma_f = \sqrt{\frac{1}{N_w T} \sum_{i=1}^{N_w T} (x_{i,f} - \mu_f)^2}$$

These global statistics describe the distribution of every joint coordinate across the entire dataset, providing a consistent normalization basis.

D. Z-Score Transformation

Once μ and σ are obtained, `StandardScaler` applies the Z-score transformation element-wise:

$$\tilde{x}_{i,f} = \frac{x_{i,f} - \mu_f}{\sigma_f}.$$

This produces a normalized representation in which every feature dimension has mean 0 and variance 1, ensuring uniform scaling across all joints and time steps.

E. Reshaping and Reproducibility

After normalization, the data is reshaped back to its 3-dimensional structure:

$$\tilde{X} \in \mathbb{R}^{N_w \times T \times F}.$$

To guarantee identical preprocessing during inference, the **StandardScaler object is saved**, preserving the computed values of μ and σ . This ensures that all new Kinect skeleton sequences follow the same normalization scheme before being passed into the MS-GRS-BiLSTM model.

3.2.4 Label Encoding and One-Hot Transformation

A. Purpose of Label Transformation

To integrate the activity labels into a supervised learning framework, the categorical activity names must be converted into numerical form. This conversion ensures full compatibility with the MS-GRS-BiLSTM architecture and allows the model to optimize using categorical loss

functions. The transformation is carried out in two stages—label encoding and one-hot encoding—to preserve class separability while maintaining computational efficiency.

B. Label Encoding (String-to-Integer Mapping)

The activity annotations are originally available as textual descriptors such as *walking*, *running*, or *sitting*. These labels are first converted into integer identifiers using a LabelEncoder, resulting in a compact numeric representation:

$$y_{\text{int}} = \text{LabelEncoder}(y_{\text{str}}).$$

For a dataset with C distinct activity categories, the encoded labels fall within the range:

$$y_{\text{int}} \in \{0, 1, \dots, C - 1\}.$$

This integer form serves as the prerequisite for constructing the one-hot target vectors required during classification.

C. One-Hot Transformation (Integer-to-Binary Class Vectors)

To facilitate optimization using the categorical cross-entropy loss, the integer-encoded labels are subsequently transformed into one-hot vectors using functions such as `to_categorical`. Each label is expanded into a binary vector of length C , where only the element corresponding to the correct activity class is assigned the value 1:

$$Y \in \{0, 1\}^{N \times C},$$

with N denoting the total number of window samples and C representing the number of activity classes. For a given sample i , the one-hot vector Y_i satisfies $Y_{i,c} = 1$ if the sample belongs to class c ; all remaining entries are 0. This structure enables stable gradient flow and well-defined decision boundaries during multi-class training.

D. Final Target Representation

Following these transformations, the training and testing target sets are expressed as:

$$y \in \mathbb{R}^{N \times C},$$

providing a standardized label representation fully aligned with the MS-GRS-BiLSTM classification pipeline.

3.2.5 Train–Test Split

A. Objective of Dataset Partitioning

Following preprocessing—standardization, windowing, and label encoding—the dataset is divided into training and testing subsets to ensure an unbiased evaluation of the MS-GRS-BiLSTM model. This separation allows performance metrics to reflect the model’s true generalization ability on unseen motion sequences.

B. Partitioning Procedure

The complete windowed dataset is represented as

$$X \in \mathbb{R}^{N_w \times T \times F}, Y \in \{0,1\}^{N_w \times C}.$$

An **80–20 stratified train–test split** is applied, preserving the proportional distribution of activity classes across both subsets. This yields $X_{\text{train}}, y_{\text{train}}$ for model learning and $X_{\text{test}}, y_{\text{test}}$ for final evaluation.

C. Final Dataset Shapes

With a fixed window length of $T = 64$, the resulting partitions have the form:

$$\begin{aligned} X_{\text{train}} &\in \mathbb{R}^{N_{\text{train}} \times 64 \times F}, y_{\text{train}} \in \mathbb{R}^{N_{\text{train}} \times C}, \\ X_{\text{test}} &\in \mathbb{R}^{N_{\text{test}} \times 64 \times F}, y_{\text{test}} \in \mathbb{R}^{N_{\text{test}} \times C}. \end{aligned}$$

This stratified 80–20 split ensures balanced class representation and supports a methodologically robust evaluation of the proposed MS-GRS-BiLSTM framework.

3.3 MS-GRS-BiLSTM Model Architecture

The MS-GRS-BiLSTM architecture is developed to effectively model the multi-scale temporal patterns and long-range dependencies inherent in Kinect-based skeletal motion data. By combining multi-scale temporal feature extraction with gated residual skip pathways and bidirectional recurrent processing, the model is able to preserve fine-grained joint dynamics while simultaneously capturing broader temporal structure. This integrated design enables the network to learn a stable and highly discriminative representation of human activities, supporting reliable recognition across diverse motion sequences.

3.3.1 Multi-Scale Temporal Learning

The Multi-Scale Temporal Learning module in the MS-GRS-BiLSTM model is designed to capture the wide range of temporal dynamics present in human activities recorded with the Kinect sensor. Human movements occur at different speeds and durations—from rapid hand

or finger gestures to longer cycles like walking or sit-to-stand transitions. Modeling these patterns with a single temporal resolution is insufficient, so we process the input sequences at multiple scales in parallel to effectively capture short-term, medium-term, and long-term motions.

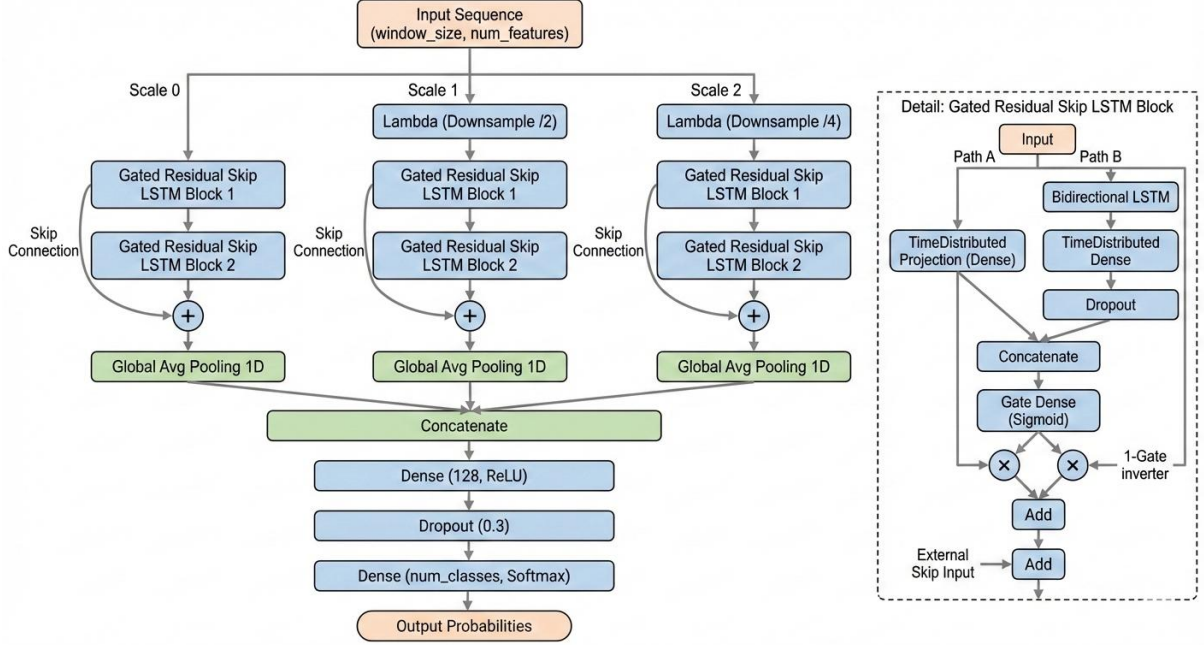


Figure 3. MS-GRS-BiLSTM model architecture

A. Motivation

Human activities naturally vary in duration. Quick motions such as finger movements or hand gestures occur within a few frames, intermediate actions like arm swings or steps span tens of frames, and longer activities like walking cycles or sit-to-stand motions occur over hundreds of frames. Using a multi-scale approach allows the model to preserve information across all these ranges, capturing both fine and global temporal patterns that a single-scale LSTM might miss.

B. Input Representation

The input to this module is the standardized, windowed skeletal sequence:

$$X \in \mathbb{R}^{B \times T \times F},$$

where B is the batch size, $T = 64$ is the number of timesteps per window, and F is the feature dimension representing the joint coordinates (e.g., 75). This provides a uniform temporal representation from which we can derive multi-scale streams.

C. Multi-Scale Stream Generation

To model motion at different temporal resolutions, we generate three streams by downsampling the input sequence. The fine-scale stream keeps all frames ($S_0 = X, T_0 = 64$) to capture rapid, short-term motions. The medium-scale stream samples every second frame ($S_1 = X[:, ::2, :], T_1 = 32$), focusing on intermediate motions while reducing redundancy. The coarse-scale stream samples every fourth frame ($S_2 = X[:, ::4, :], T_2 = 16$), highlighting long-term trends. Each stream retains the full feature dimension F , ensuring that spatial information of the skeleton joints is preserved. This downsampling allows the model to focus on relevant temporal dependencies at each scale while also reducing computational complexity.

D. Multi-Scale Processing

Each stream is processed independently by a dedicated GRS-BiLSTM block, which learns temporal dependencies specific to that resolution. The outputs from all scales are then fused to create a unified spatio-temporal representation. This combined representation captures both subtle, fast skeletal changes and extended motion patterns, improving the MS-GRS-BiLSTM model's ability to accurately recognize complex human activities.

3.3.2 Bidirectional LSTM with Gated Residual Skip Connections

The GRS-BiLSTM block serves as the core processing unit in the MS-GRS-BiLSTM architecture. It transforms multi-scale temporal sequences into rich, discriminative features by sequentially combining LSTM memory, bidirectional temporal context, gated fusion, and residual skip connections. This design ensures both short-term and long-term skeletal motion patterns are effectively captured, which is essential for recognizing complex human activities.

A. Input and Initial Projection

The input to a GRS-BiLSTM block is a downsampled sequence:

$$x_{in} \in \mathbb{R}^{B \times T_i \times F_{in}}$$

where B is the batch size, T_i is the number of timesteps at the current scale, and F_{in} is the feature dimension. The first step is a **residual projection**, which linearly maps the input into the hidden dimension U while preserving local frame-wise motion features:

$$td_proj(t) = W_p x_{in}(t) + b_p$$

Here, $W_p \in \mathbb{R}^{F_{in} \times U}$ and $b_p \in \mathbb{R}^U$. This residual path bypasses temporal modeling, ensuring fine-grained motion information is preserved for later fusion with LSTM outputs.

B. LSTM-Based Temporal Encoding

The projected sequence is then processed by an LSTM to capture sequential dependencies through its memory cell C_t and hidden state h_t :

$$\begin{aligned}
 f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) && \text{(Forget Gate)} \\
 i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i) && \text{(Input Gate)} \\
 \tilde{C}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c) && \text{(Candidate Cell State)} \\
 C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t && \text{(Cell State Update)} \\
 o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), h_t = o_t \odot \tanh(C_t) && \text{(Output Gate and Hidden State)}
 \end{aligned}$$

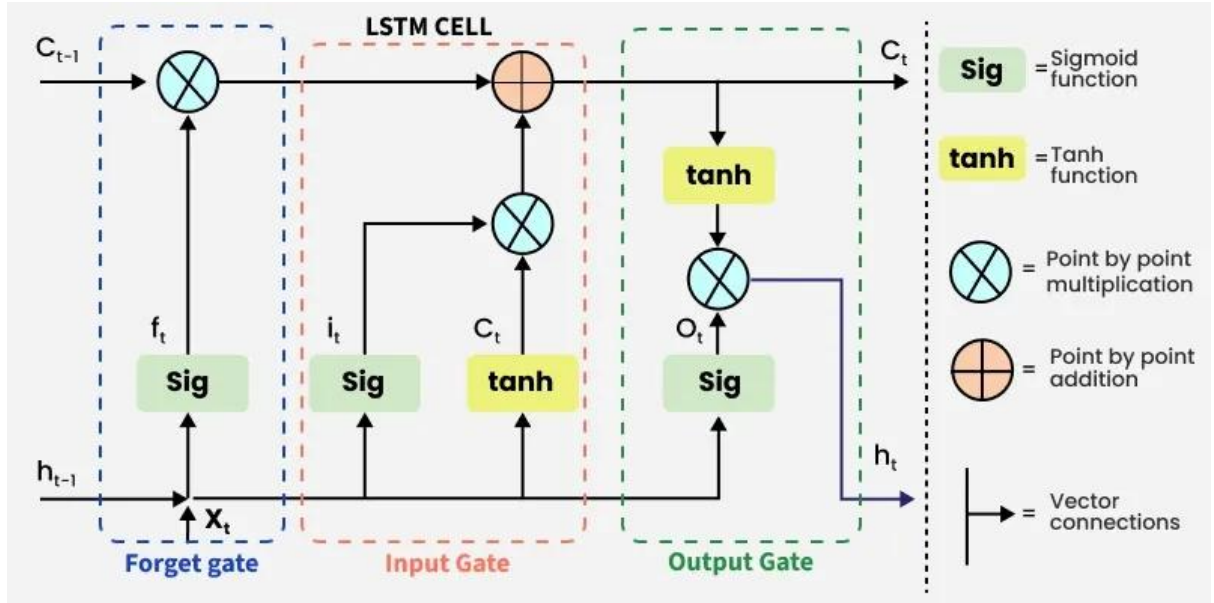


Figure 4. LSTM model architecture

The LSTM retains short-term fluctuations and long-term dependencies in skeletal motions, which is critical for capturing micro-movements and extended activity cycles.

C. Bidirectional Contextual Learning

To further enhance temporal representation, the LSTM is extended into a **Bidirectional LSTM (BiLSTM)** to incorporate information from both past and future frames:

$$\vec{h}_t = \text{LSTM}_{forward}(x_t), \tilde{h}_t = \text{LSTM}_{backward}(x_t) \quad \text{(Forward and Backward LSTM)}$$

$$h_t = [\vec{h}_t; \tilde{h}_t] \in \mathbb{R}^{2U} \quad \text{(Concatenated Hidden State)}$$

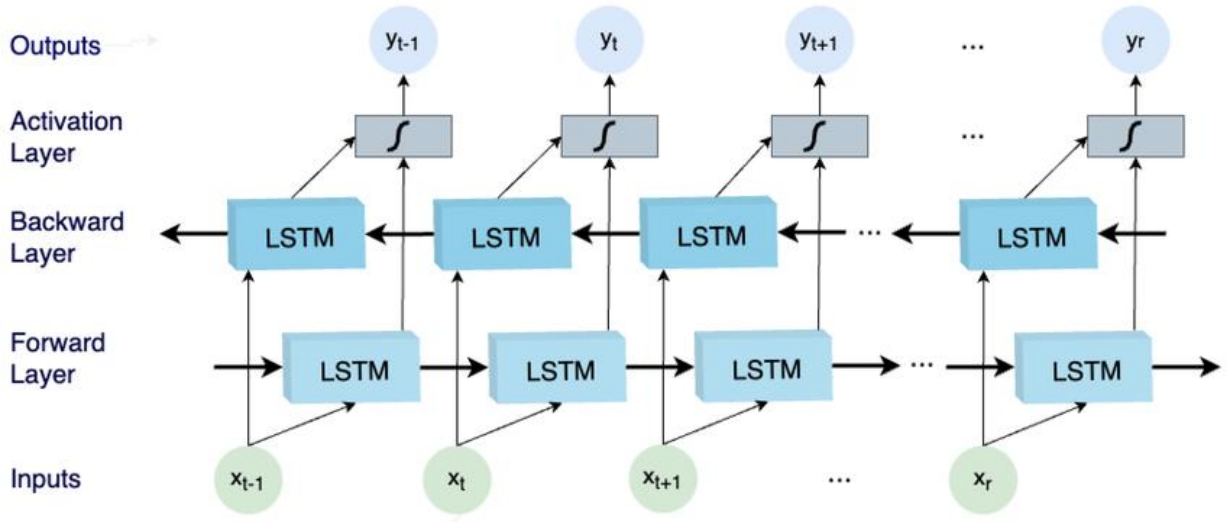


Figure 5. BiLSTM model architecture

Concatenating forward and backward hidden states ensures each timestep contains context-aware temporal information, which improves recognition of both anticipatory and sequential activity patterns.

D. Gated Residual Fusion

The BiLSTM output $\tilde{h}_t$ is fused with the residual projection $td_proj(t)$ through a **learned gate** to adaptively combine temporal and local features:

$$z_t = [td_proj(t); \tilde{h}_t], g_t = \sigma(W_g z_t + b_g)$$

$$out(t) = g_t \odot \tilde{h}_t + (1 - g_t) \odot td_proj(t)$$

The gate $g_t \in (0,1)^U$ allows the network to selectively emphasize either long-term temporal dynamics ($\tilde{h}_t$) or fine-grained frame-level motions ($td_proj(t)$) depending on the activity being recognized.

E. Residual Skip Across Blocks

For deep temporal modeling, multiple GRS-BiLSTM blocks are stacked. A **residual skip connection** adds the output from the previous block directly to the current block output:

$$out_{final} = out + skip_connection$$

This skip connection preserves features across layers, stabilizes gradient flow, mitigates vanishing gradients, and accelerates convergence. It ensures that early-stage and higher-level

temporal features are effectively utilized throughout the network, improving the robustness and accuracy of activity recognition.

3.3.3 Temporal Feature Aggregation and Fusion

The Temporal Feature Aggregation and Fusion module in the MS-GRS-BiLSTM architecture consolidates the temporal representations learned at multiple scales into a unified feature vector suitable for activity classification. This process ensures that both short-term and long-term skeletal motion patterns contribute to the final representation.

A. Temporal Global Pooling per Scale

For each temporal scale $i \in \{0,1,2\}$, the output of the second GRS-BiLSTM block $b_{i,2} \in \mathbb{R}^{B \times T_i \times U}$ is aggregated across the time dimension. Here, B is the batch size, T_i is the number of timesteps at scale i , and U is the number of LSTM units per direction.

Global Average Pooling (GAP) is applied to summarize the temporal dynamics into a fixed-size feature vector:

$$p_i = \text{GAP}(b_{i,2}) = \frac{1}{T_i} \sum_{t=1}^{T_i} b_{i,2}(:, t, :) \quad (\text{Temporal Global Average Pooling})$$

The resulting vector $p_i \in \mathbb{R}^{B \times U}$ captures the dominant activity patterns at the corresponding temporal scale while reducing redundancy and computational complexity. This aggregation allows the model to retain essential motion features from short-term gestures, medium-range movements, and long-term actions.

B. Multi-Scale Feature Fusion

After pooling, the feature vectors from all three scales— p_0 (fine/full), p_1 (medium/stride 2), and p_2 (coarse/stride 4)—are concatenated to form a unified representation:

$$p = [p_0, p_1, p_2] \in \mathbb{R}^{B \times 3U} \quad (\text{Multi-Scale Feature Concatenation})$$

This concatenation integrates complementary temporal information across scales, allowing the network to leverage short-term, medium-term, and long-term dependencies simultaneously. The fused feature vector p serves as a comprehensive representation of the human activity, ready to be passed to the output classification layer.

By combining multi-scale temporal features through global pooling and concatenation, this module ensures that the MS-GRS-BiLSTM model captures a holistic view of skeletal motion dynamics, enhancing its discriminative power for complex activity recognition tasks.

3.3.4 Output Classification Layer

Once the multi-scale temporal features have been aggregated and fused into the final representation $p \in \mathbb{R}^{B \times 3U}$, the model transitions into the classification stage. At this point, my goal is to transform the rich temporal descriptors extracted from the three scales into meaningful activity predictions.

A. Fusion Dense Transformation

I begin by passing the fused feature vector p through a fully connected layer. This step helps the model integrate the complementary information learned from the full-resolution, medium-resolution, and highly downsampled temporal streams. The transformation is computed as:

$$f = \phi(W_f p + b_f)$$

Here, $W_f \in \mathbb{R}^{3U \times 128}$ is the weight matrix, $b_f \in \mathbb{R}^{128}$ is the bias term, and $\phi(\cdot)$ is the ReLU activation. The resulting feature vector $f \in \mathbb{R}^{B \times 128}$ captures a compact and discriminative representation of the activity dynamics. The parameter count for this layer is:

$$P_{\text{fusion}} = (3U) \cdot 128 + 128$$

B. Final Softmax Classification

After obtaining the fused and transformed feature vector, I use it to generate the final activity predictions. The regularized output f' from the previous step (after dropout) is passed into the final Dense layer, which uses the softmax activation to produce class probabilities:

$$\hat{y} = \text{softmax}(W_c f' + b_c)$$

where $W_c \in \mathbb{R}^{128 \times C}$ and $b_c \in \mathbb{R}^C$. The vector $\hat{y} \in \mathbb{R}^{B \times C}$ contains the predicted probability for each of the C activities. The parameter count for this classifier layer is:

$$P_{\text{classifier}} = 128 \cdot C + C$$

For example, with eight activity classes, this becomes $128 \times 8 + 8 = 1032$.

The model is trained using categorical cross-entropy, and the gradient entering the backward pass from the softmax layer is:

$$\frac{\partial L}{\partial z} = \hat{y} - y$$

4. Experiments

4.1 Experimental Setup

The experimental setup for this Human Activity Recognition (HAR) study was designed to acquire accurate and reliable human motion data using a non-intrusive sensing framework. Data collection was carried out using the **Microsoft Kinect v2 sensor**, which provides real-time multimodal information including RGB imagery, depth maps, and three-dimensional skeletal joint positions. The contact-free nature of the sensor enables participants to perform activities naturally without constraints imposed by wearable devices.

The Kinect v2 sensor tracks a **25-joint skeletal model**, with each joint represented by three-dimensional spatial coordinates (x, y, z) relative to the sensor reference frame. All joint positions are expressed in meters, allowing precise representation of human posture and movement. Skeletal data were captured at an approximate frame rate of **30 Hz**, which is sufficient to accurately model both slow and dynamic human actions.

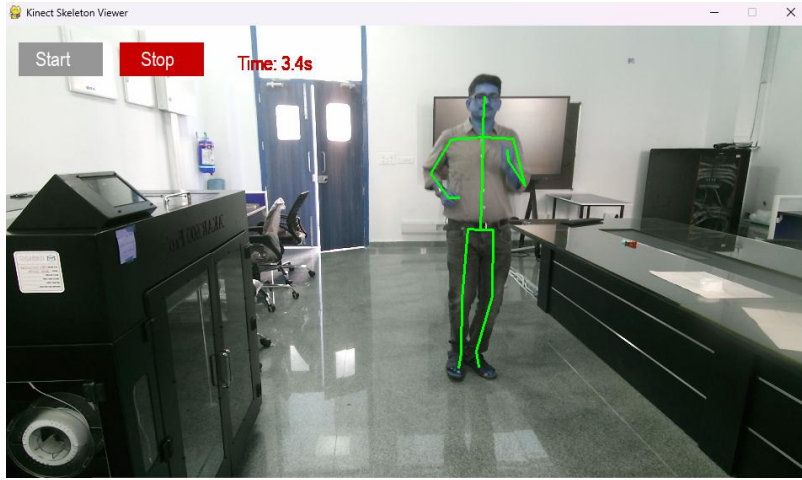


Figure 6. Volunteer performing activities

A **custom skeletal dataset** was collected specifically for this study. The dataset consists of recordings from **30 volunteers** of varying ages, body shapes, heights, and genders to capture inter-subject motion variability. Each participant performed **seven daily activities**: bending, jumping, running, sitting, squatting, standing, and walking. Data collection was conducted in a controlled indoor environment to reduce external disturbances while preserving natural movement execution. The dataset is organized hierarchically by subject, with individual folders containing separate CSV files for each activity. Each CSV file stores time-series skeletal data, where each frame contains the 3D coordinates of all tracked joints along with auxiliary metadata.

To assess the general applicability of the experimental setup across datasets, the **Kinect Activity Recognition Dataset (KARD)** was employed as an external benchmark. KARD

consists of skeletal recordings of **18 human activities** performed three times by **10 subjects**, resulting in a total of 540 sequences. The activities included in KARD are: **horizontal arm wave, high arm wave, two hand wave, catch cap, high throw, draw X, draw tick, toss paper, forward kick, side kick, take umbrella, bend, hand clap, walk, phone call, drink, sit down, and stand up**. These activities span upper-body, lower-body, and full-body movements, providing a comprehensive evaluation of varied motion patterns.

Before experimentation, all skeletal data were subjected to a consistent preprocessing procedure. Non-essential attributes such as timestamps and body identifiers were removed, retaining only the 3D joint coordinates. Missing or corrupted joint values were handled using forward-fill and backward-fill strategies, and any remaining gaps were replaced with zeros to maintain temporal continuity. This preprocessing ensured clean and uniform skeletal representations while preserving the original motion characteristics of each activity.

4.2 Evaluation Protocols

To comprehensively evaluate the performance of the human activity recognition system, multiple standard classification metrics were employed. These metrics were selected to provide a balanced assessment of predictive accuracy, class-wise discrimination capability, and error characteristics, particularly in the presence of inter-class similarity and potential class imbalance commonly observed in HAR datasets.

Precision measures the proportion of correctly predicted positive samples out of all samples predicted as positive. It reflects the reliability of positive predictions and is especially important in scenarios where false positives are costly, such as incorrectly identifying a critical activity when none has occurred. Precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Recall quantifies the proportion of correctly predicted positive samples out of all actual positive samples. This metric is crucial when missing a true activity instance is undesirable, such as failing to detect an actual event of interest. Recall is computed as:

$$Recall = \frac{TP}{TP + FN}$$

F1-score is the harmonic mean of precision and recall and provides a single balanced measure that accounts for both false positives and false negatives. This metric is particularly effective for evaluating performance on imbalanced activity classes. The F1-score is defined as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Accuracy represents the proportion of correctly classified samples out of the total number of samples and serves as a general indicator of overall classification performance. Although widely used, accuracy may be less reliable in the presence of class imbalance. Accuracy is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

In addition to classification metrics, **loss** was monitored to evaluate prediction error and guide the optimization process. The **categorical cross-entropy loss** function was used to measure the discrepancy between predicted class probabilities and ground-truth labels. It is defined as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$$

where (N) denotes the total number of samples, (C) represents the number of activity classes, ($y_{i,c}$) is the ground-truth label (equal to 1 if sample (i) belongs to class (c), and 0 otherwise), and ($p_{i,c}$) is the predicted probability of class (c) for sample.

In all equations, **TP** denotes true positives (correctly predicted positive samples), **FP** represents false positives (incorrect positive predictions), **FN** corresponds to false negatives (missed actual positives), and **TN** indicates true negatives (correct negative predictions). Together, these evaluation protocols provide a robust and comprehensive framework for assessing the performance of human activity recognition systems across multiple activities and experimental conditions.

5. Results and Discussion

5.1 Classification Accuracy

This section presents a detailed analysis of the classification performance achieved by the proposed **Multi-Scale Gated Residual Skip BiLSTM (MS-GRS-BiLSTM)** model in comparison with baseline recurrent architectures. The discussion not only reports numerical results but also explains the underlying reasons for the observed performance trends, highlighting both strengths and limitations of the evaluated approaches.

The classification accuracy obtained on the custom skeletal joint dataset is summarized in **Table 4**. The standard LSTM model achieves an accuracy of **76.19%**, which can be attributed to its unidirectional temporal processing that restricts the model to learning only past dependencies. Human activities, however, often involve transitional and anticipatory motion

patterns, which are not fully captured by such a formulation. The BiLSTM model improves the accuracy to **88.10%** by incorporating both forward and backward temporal contexts, enabling a better understanding of overlapping and sequential movements. The GRS-BiLSTM further enhances performance to **93.33%** by introducing gated residual skip connections that facilitate stable gradient flow and preserve discriminative skeletal features across deeper layers. The proposed MS-GRS-BiLSTM achieves the highest accuracy of **98.35%**, demonstrating that multi-scale temporal processing effectively captures both rapid limb movements and slow postural transitions, which are common in daily human activities.

Table 4. Classification Accuracy Comparison on the Custom Skeletal Joint Dataset

Model	Temporal Modeling	Architectural Features	Test Accuracy (%)
LSTM	Unidirectional	Basic sequential modeling	76.19
BiLSTM	Bidirectional	Past and future context learning	88.10
GRS-BiLSTM	Bidirectional	Gated residual skip connections	93.33
MS-GRS-BiLSTM (Proposed)	Multi-scale + Bidirectional	Multi-scale temporal learning with gated residual skips	98.35

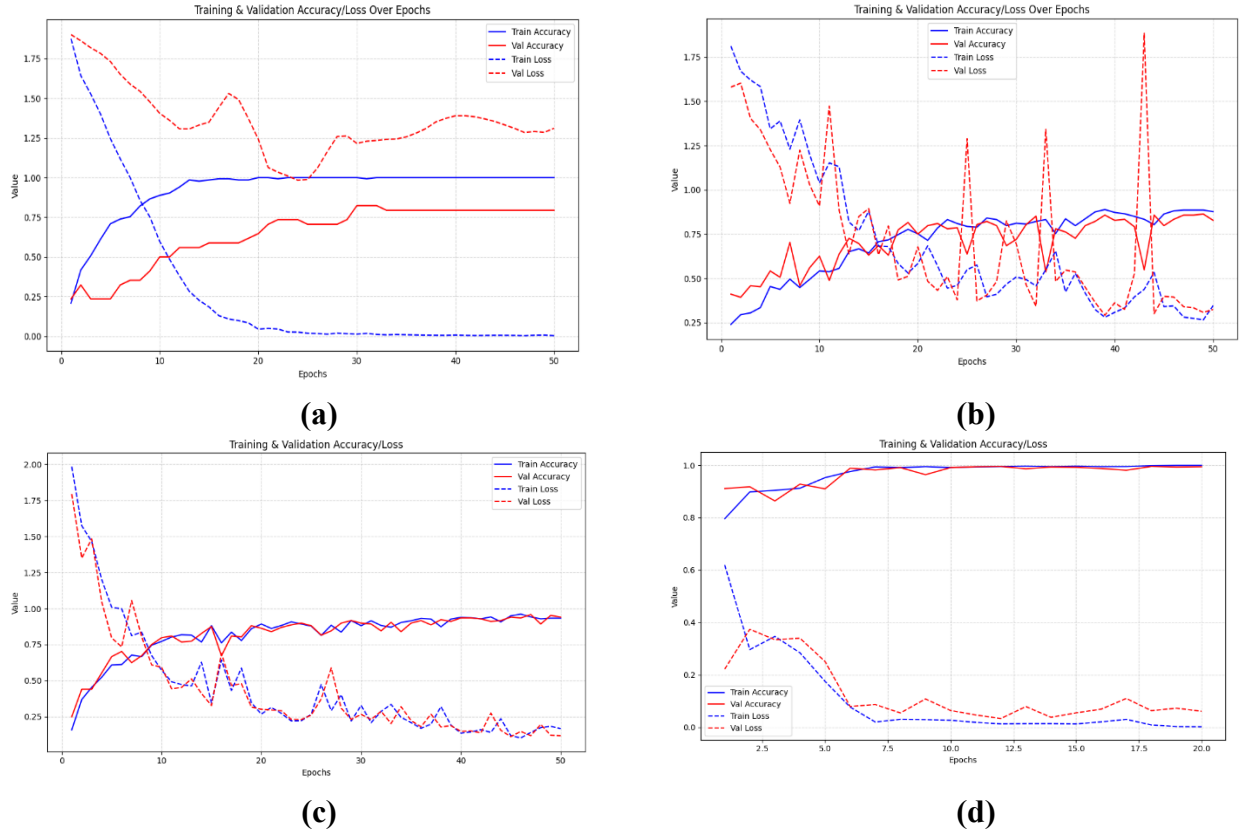


Figure 7. Accuracy & Loss over training and validation over Custom dataset using (a) LSTM (b) BiLSTM (c) GRS-BiLSTM (d) MS-GRS-BiLSTM model

To further understand the reliability of the proposed approach, precision, recall, and F1-score were analyzed and are reported in **Table 5**. The MS-GRS-BiLSTM achieves a weighted F1-score of **0.9835**, indicating a strong balance between false positives and false negatives across activity classes. This balanced performance confirms that the model does not overfit dominant activities and maintains consistent recognition accuracy even for less frequent actions.

Table 5 Detailed Performance Metrics of MS-GRS-BiLSTM on the Custom Dataset

Metric	Value
Precision (Weighted Avg)	0.9836
Recall (Weighted Avg)	0.9835
F1-Score (Weighted Avg)	0.9835
Overall Accuracy (%)	98.35

A class-wise analysis of F1-scores is presented in **Table 6**. Activities such as **sitting** and **standing** achieve perfect F1-scores due to their stable and well-defined skeletal configurations,

which exhibit minimal intra-class variation. Slow dynamic actions such as **squatting** and **walking** also achieve near-perfect performance, benefiting from consistent joint trajectories. In contrast, fast and highly dynamic actions such as **jumping** and **running** exhibit slightly lower F1-scores, primarily due to increased inter-subject variability, rapid joint acceleration, and occasional self-occlusions. Nevertheless, the performance remains high, demonstrating the robustness of the proposed model.

Table 6. Activity-wise F1-Score Analysis on the Custom Dataset (MS-GRS-BiLSTM)

Activity	Motion Type	F1-Score
Sitting	Static posture	1.00
Standing	Static posture	1.00
Squatting	Slow dynamic	0.99
Walking	Cyclic motion	0.99
Bending	Moderate dynamic	0.98
Jumping	Fast dynamic	0.94
Running	Fast dynamic	0.94

To evaluate generalization capability, the models were further tested on the public **Kinect Activity Recognition Dataset (KARD)**, which includes **18 diverse activities** involving both upper-body and full-body motions. The accuracy comparison is reported in **Table 7**. Similar trends are observed, with the proposed MS-GRS-BiLSTM achieving a benchmark-leading accuracy of **98.77%**. This strong performance indicates that the learned multi-scale temporal representations are not dataset-specific and generalize well to unseen activity patterns and subjects.

Table 7. Classification Accuracy Comparison on the KARD Dataset

Model	Dataset Complexity	Test Accuracy (%)
LSTM	Multi-class (18 activities)	77.78
BiLSTM	Multi-class (18 activities)	87.65
GRS-BiLSTM	Multi-class (18 activities)	92.59
MS-GRS-BiLSTM (Proposed)	Multi-class (18 activities)	98.77

The detailed macro-averaged metrics for the KARD dataset are provided in **Table 8**. The proposed model achieves a macro F1-score of **0.9870**, confirming balanced performance across all 18 activities. Most actions achieve perfect precision, indicating an extremely low false-

positive rate. Minor recall variations are observed in actions such as **Catch Cap** and **Phone Call**, which involve subtle hand movements and limited skeletal displacement, making them inherently challenging for Kinect-based skeleton tracking.

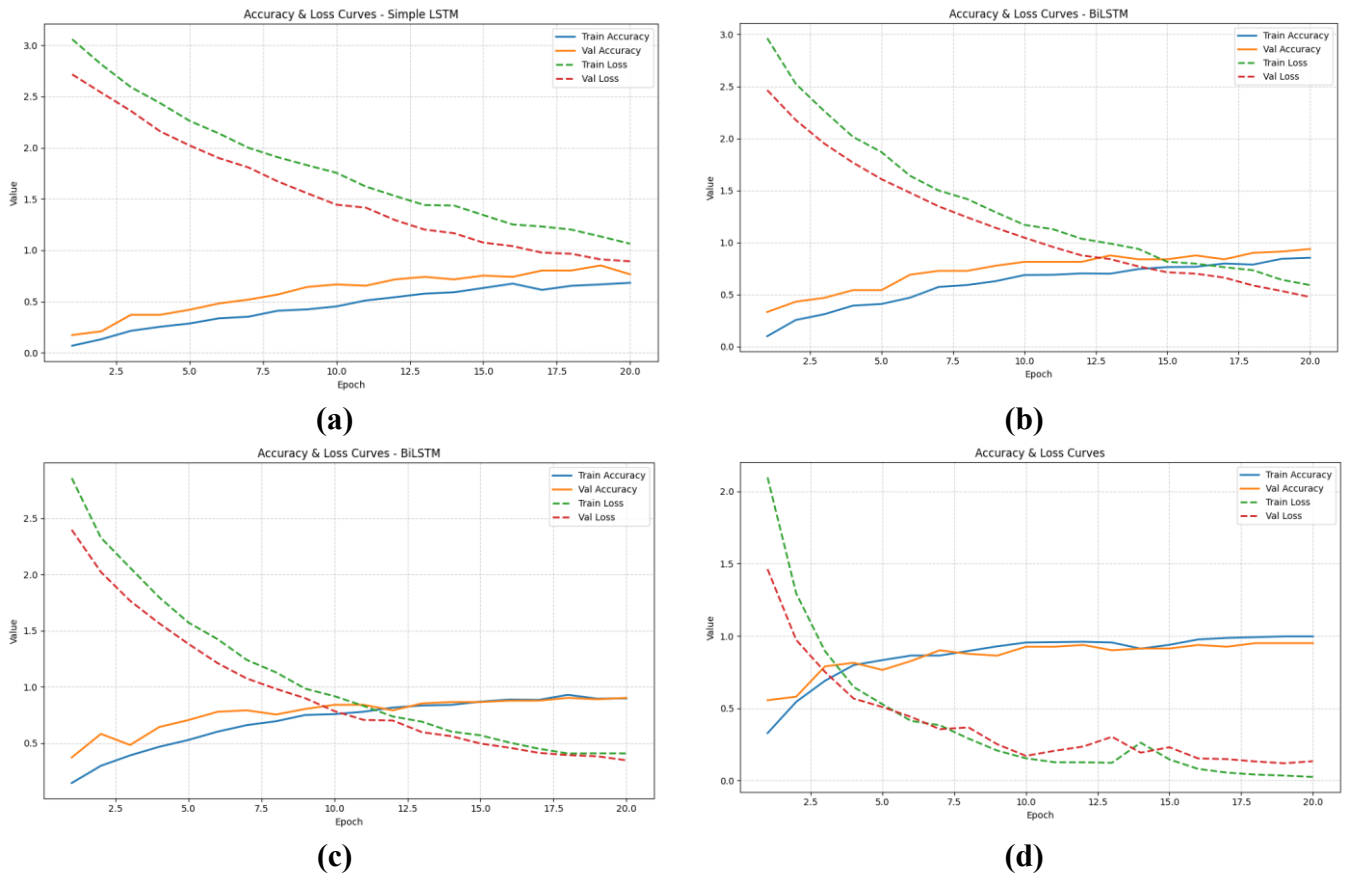


Figure 8. Accuracy & Loss over training and validation on KARD dataset using (a) LSTM (b) BiLSTM (c) GRS-BiLSTM (d) MS-GRS-BiLSTM model

Table 8. Detailed Performance Metrics of MS-GRS-BiLSTM on the KARD Dataset

Metric	Macro Average Value
Precision	0.9907
Recall	0.9861
F1-Score	0.9870
Overall Accuracy (%)	98.77

A summary of action-level behavior on the KARD dataset is presented in **Table 9**, highlighting that 16 out of 18 activities achieve perfect precision. The remaining two actions exhibit minor

recall degradation due to fine-grained motion characteristics and occasional occlusions, which represent an inherent limitation of vision-based skeletal sensing.

Table 9. Action-wise Precision Summary on KARD Dataset

Category	Number of Actions
Actions with Precision = 1.0000	16
Actions with Minor Recall Variations	2
Most Challenging Actions	Catch Cap, Phone Call

Finally, computational efficiency is evaluated in **Table 10**, which compares accuracy with inference time and model size. Despite achieving the highest accuracy, the proposed MS-GRS-BiLSTM records an average inference time of only **1.857 ms per sample** and a compact model size of **6.82 MB**. This efficiency is attributed to effective multi-scale feature reuse and gated residual pathways, which reduce redundant computations while preserving discriminative temporal information. These characteristics make the proposed model particularly suitable for real-time human activity recognition and robot imitation applications.

Table 10. Comparative Summary of Accuracy and Computational Efficiency

Model	Custom Dataset Accuracy (%)	KARD Accuracy (%)	Avg. Inference Time (ms/sample)	Model Size (MB)
LSTM	76.19	77.78	~31.01	Larger
BiLSTM	88.10	87.65	~27.03	Moderate
GRS-BiLSTM	93.33	92.59	~31.86	Moderate
MS-GRS-BiLSTM (Proposed)	98.35	98.77	1.857	6.82

Overall, the results demonstrate that the proposed MS-GRS-BiLSTM effectively addresses the challenges of temporal modeling in skeletal-based human activity recognition. While the model exhibits exceptional accuracy and real-time efficiency, its performance remains dependent on reliable skeletal tracking, and subtle hand-centric actions continue to pose challenges. These observations provide clear directions for future research, including enhanced joint modeling and multi-sensor fusion.

5.2 Confusion Matrix Analysis

Confusion matrix analysis provides detailed insight into class-wise prediction behavior by highlighting both correct classifications and systematic misclassifications across activity classes. Unlike overall accuracy, confusion matrices expose the underlying reasons for model success or failure by revealing inter-class similarities, temporal overlap, and structural limitations in motion representation. This section presents a comprehensive confusion matrix analysis for both the custom skeletal joint dataset and the Kinect Activity Recognition Dataset (KARD), followed by a comparative discussion of observed trends.

5.2.1 Custom Skeletal Joint Dataset

The confusion matrices obtained on the custom skeletal joint dataset, consisting of seven activities, demonstrate a clear progression in class separability as the modeling framework evolves from conventional recurrent architectures to the proposed MS-GRS-BiLSTM. The proposed model exhibits a near-ideal diagonal structure, indicating minimal misclassification across all activity classes. Static postures such as **Sitting** and **Standing** are recognized with perfect accuracy, with all instances correctly classified. This strong performance can be attributed to the consistent skeletal configurations and limited temporal variability associated with these activities.

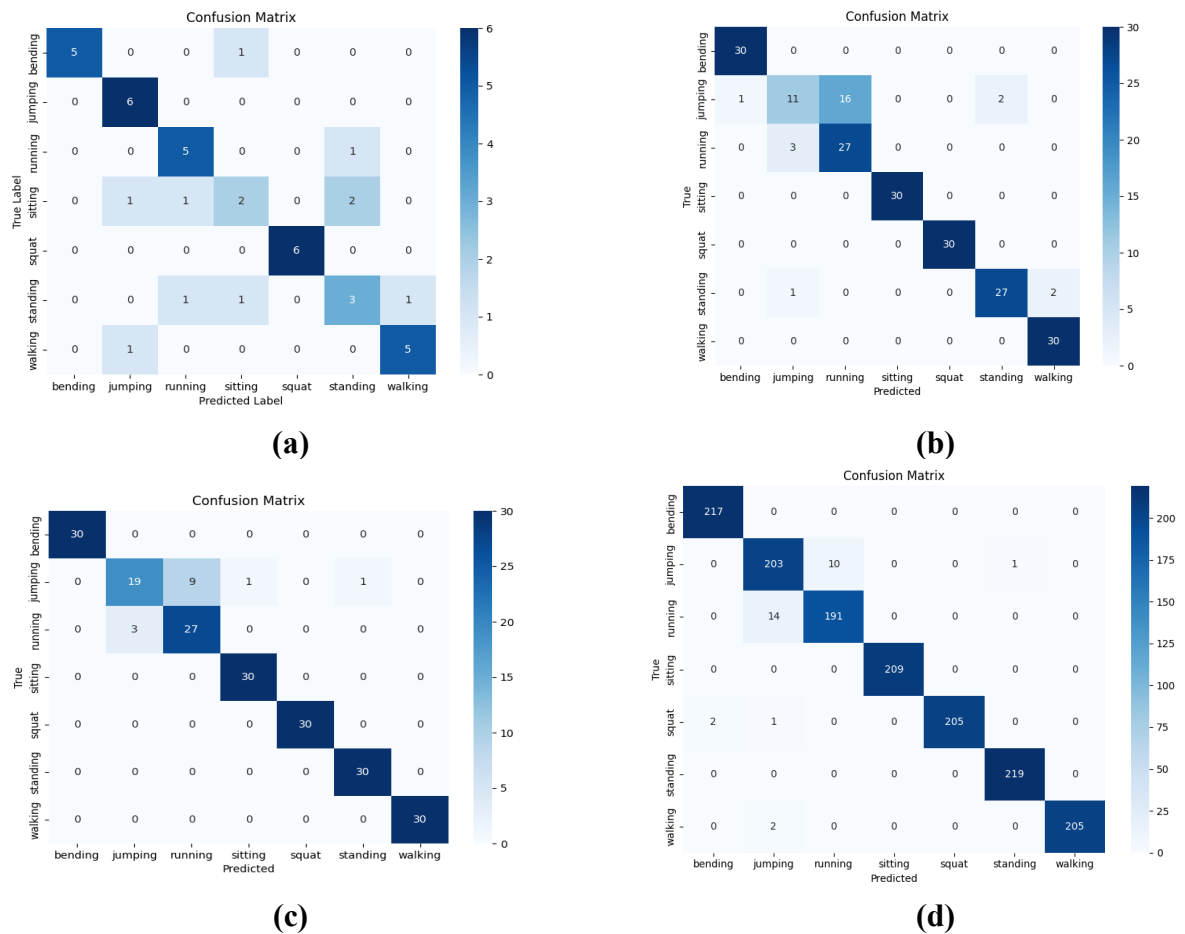


Figure 9. Confusion Matrix on Custom Dataset using MS-GRS-BiLSTM model using (a) LSTM (b) BiLSTM (c) GRS-BiLSTM (d) MS-GRS-BiLSTM mode

Despite its overall robustness, the proposed model shows minor confusion between **Jumping** and **Running**, where a small number of samples are mutually misclassified. This confusion arises from biomechanical similarities such as rapid lower-limb motion, overlapping joint velocity profiles, and comparable center-of-mass trajectories. Additionally, a very small number of **Squatting** samples are misclassified as **Bending** or **Jumping**, which reflects transitional postures where joint angles and body orientation momentarily resemble adjacent actions. However, these misclassifications remain limited in number and do not significantly affect overall performance. In contrast, the **GRS-BiLSTM** shows increased off-diagonal dispersion for dynamic activities. While it successfully classifies static and moderately dynamic actions such as bending, squatting, and walking, its performance deteriorates for jumping and running. This indicates that although gated residual connections help preserve spatial skeletal information across layers, the lack of explicit multi-scale temporal modeling limits its ability to disambiguate actions that differ primarily in speed and motion frequency.

The **BiLSTM** further amplifies this limitation, as it struggles to maintain discriminative temporal features for rapid actions, leading to substantial confusion in jumping-related movements. The baseline **LSTM** exhibits the most dispersed confusion matrix, particularly failing to distinguish between sitting and standing. This outcome highlights the inability of unidirectional temporal processing to effectively separate activities with similar static joint configurations but subtle temporal differences.

5.2.2 Kinect Activity Recognition Dataset (KARD)

The confusion matrix analysis on the KARD dataset, which contains 18 diverse activities involving complex upper-body and full-body motions, further validates the generalization capability of the proposed MS-GRS-BiLSTM architecture. The confusion matrix for the proposed model is characterized by a strong diagonal dominance, with 16 out of 18 activities achieving perfect precision. This result indicates that the model produces extremely few false positives across a wide range of motion patterns.

The only notable confusion occurs between **Action_15 (Phone Call)** and **Action_4 (Catch Cap)**, where a single instance of the phone call action is misclassified. This confusion is primarily due to the similarity in arm elevation and hand-to-head movement patterns, compounded by the limited resolution of skeletal joint data for fine-grained hand gestures. Nevertheless, the rarity of such errors underscores the robustness of the proposed approach when applied to complex, multi-class activity recognition problems.

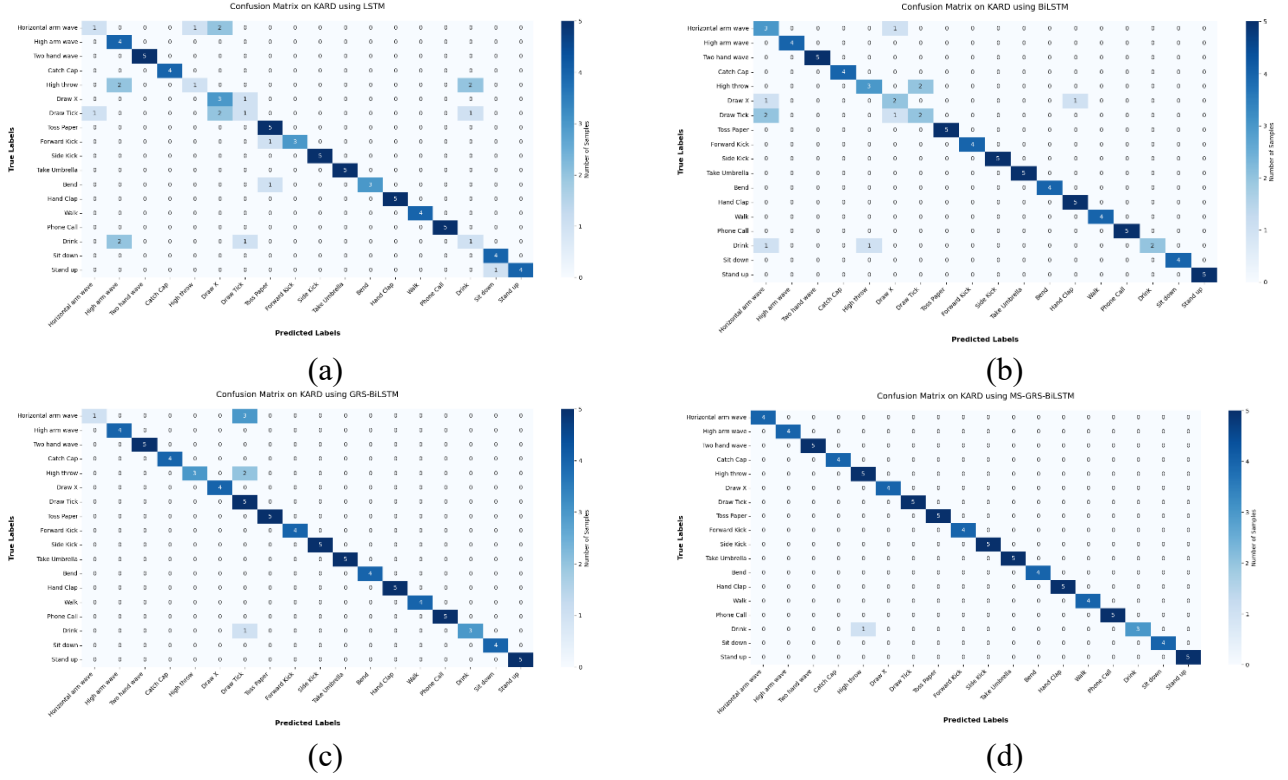


Figure 10. Confusion Matrix on KARD Dataset using MS-GRS-BiLSTM model using (a) LSTM (b) BiLSTM (c) GRS-BiLSTM (d) MS-GRS-BiLSTM model

The **GRS-BiLSTM** demonstrates reasonable performance on the KARD dataset but exhibits distinct confusion clusters for specific actions. Certain activities suffer from low recall or precision, particularly those involving subtle hand motions or short-duration gestures. These results suggest that residual connections alone are insufficient to fully resolve fine temporal variations without additional temporal abstraction. The **BiLSTM** shows a broader spread of misclassifications, with several gesture-based actions exhibiting degraded F1-scores, indicating limitations in capturing short-term temporal dependencies. The **LSTM**, as expected, produces the most populated off-diagonal entries, struggling significantly with subtle and overlapping gestures, which confirms its limited suitability for high-resolution multi-class skeletal activity recognition.

5.2.3 Cross-Model Confusion Trends and Discussion

A comparative examination of confusion matrices across all models reveals a systematic reduction in misclassification as architectural enhancements are introduced. Static posture confusion is largely mitigated by incorporating bidirectional temporal context, enabling the model to leverage both past and future skeletal information. Temporal overlap between dynamic activities, such as jumping and running, is further reduced through multi-scale temporal processing, which allows the model to distinguish between rapid bursts of motion and sustained movement patterns. Additionally, gated residual skip connections play a critical role

in preventing information degradation across deep layers, ensuring that essential skeletal cues are preserved during long temporal sequences.

Although minor misclassifications persist for activities with highly similar skeletal trajectories or minimal joint displacement, the overall confusion structure of the MS-GRS-BiLSTM remains highly organized and interpretable. The concentration of predictions along the main diagonal, combined with the sparsity of off-diagonal errors, confirms the effectiveness of the proposed architecture in capturing both spatial and temporal dynamics. These findings demonstrate that the MS-GRS-BiLSTM significantly improves class separability across diverse datasets, making it well-suited for real-time human activity recognition and robot imitation systems.

5.3 Impact of Multi-Scale and Residual Connections

The superior performance of the proposed MS-GRS-BiLSTM architecture is primarily attributed to the synergistic integration of **Multi-Scale (MS) temporal modeling** and **Gated Residual Skip (GRS) connections**. These two components are explicitly designed to address the inherent limitations of conventional recurrent neural networks when applied to skeletal-based human activity recognition, particularly in handling motion speed variability and preventing information degradation across deep temporal layers. This section analyzes the individual and combined impact of these components across both the custom skeletal joint dataset and the KARD benchmark.

5.3.1 Impact of Multi-Scale Temporal Modeling

Human activities exhibit significant variation in temporal dynamics, ranging from rapid, repetitive motions such as running and jumping to slower, posture-dominated actions such as sitting and bending. A single temporal resolution is often insufficient to capture these variations effectively, as critical motion cues may either be oversampled or lost entirely. The Multi-Scale component addresses this challenge by processing skeletal sequences at three distinct temporal resolutions: the original full-resolution sequence, a medium-resolution sequence sampled at every second frame, and a coarse-resolution sequence sampled at every fourth frame.

This multi-resolution analysis enables the model to capture fine-grained motion details at the highest temporal resolution while simultaneously modeling long-term structural trends at coarser scales. As a result, activities that appear ambiguous when observed at a single speed become more distinguishable when evaluated across multiple temporal perspectives. On the custom skeletal dataset, this capability significantly reduces confusion between dynamic actions such as jumping and running, which often share similar instantaneous joint configurations but differ in motion frequency and duration. The inclusion of Multi-Scale modeling increases the F1-score for jumping from 0.73 in the GRS-BiLSTM to 0.9469, and for running from 0.82 to 0.9467, demonstrating its effectiveness in resolving speed-related ambiguities.

The impact of Multi-Scale modeling is also evident at the dataset level. On the custom dataset, overall accuracy improves from 93.33% with the GRS-BiLSTM to 98.35% with the MS-GRS-BiLSTM. A similar trend is observed on the KARD dataset, where Multi-Scale processing enhances discrimination among subtle gesture-based actions, pushing accuracy from 92.59% to 98.77%. Despite these benefits, the Multi-Scale approach introduces additional computational pathways, which may slightly increase architectural complexity. However, this overhead is effectively mitigated by efficient parallel processing and does not adversely affect real-time inference performance.

5.3.2 Impact of Gated Residual Skip Connections

While Multi-Scale modeling enhances temporal representation, deep recurrent networks are still susceptible to information loss as data propagates through multiple layers. This issue is particularly problematic for skeletal data, where subtle joint-level details are essential for distinguishing between visually similar activities. The Gated Residual Skip (GRS) mechanism addresses this limitation by introducing a learnable pathway that directly connects the input skeletal representation to deeper layers, allowing critical spatial information to be preserved.

The GRS mechanism combines the temporally processed BiLSTM output with a projected version of the original input using a learnable gate. This gate adaptively controls the contribution of newly extracted temporal features versus preserved skeletal structure at each time step. As a result, the network avoids excessive smoothing of joint trajectories and maintains sensitivity to fine postural differences. In addition, the residual pathway improves gradient flow during backpropagation, alleviating the vanishing gradient problem and stabilizing training for deeper architectures.

The effectiveness of GRS connections is clearly observed when comparing BiLSTM and GRS-BiLSTM models. On the custom dataset, the inclusion of GRS improves accuracy from 88.10% to 93.33%, with notable gains in recognizing static postures such as sitting and standing. These activities share similar skeletal geometry and are particularly vulnerable to feature loss in standard recurrent networks. On the KARD dataset, GRS connections help resolve misclassifications in actions such as Action_0 and Action_6, where subtle hand and arm movements require preserved contextual information for accurate recognition. A potential limitation of GRS is the increased number of learnable parameters, but this cost is justified by the substantial gains in stability and representational fidelity.

5.3.3 Combined Effect of Multi-Scale and Gated Residual Design

The full benefit of the proposed architecture emerges when Multi-Scale temporal modeling and Gated Residual Skip connections are combined within the MS-GRS-BiLSTM framework. Multi-Scale processing enhances temporal diversity by analyzing motion at different speeds, while GRS connections ensure that essential skeletal information is retained throughout the

network. Together, these components form a complementary system that captures both short-term motion dynamics and long-term structural consistency.

This combined effect is reflected in the consistent performance improvements observed across both datasets. The MS-GRS-BiLSTM achieves the highest classification accuracy on the custom dataset (98.35%) and the KARD dataset (98.77%), outperforming all baseline models by a significant margin. Moreover, the architecture achieves these gains without sacrificing computational efficiency, making it well-suited for real-time human activity recognition and robot imitation tasks. Although minor confusions persist for activities involving extremely subtle gestures or overlapping motion patterns, the combined design substantially minimizes such errors and produces highly structured, interpretable predictions.

5.3.4 Comparative Impact on Classification Accuracy

The cumulative contribution of Multi-Scale and Gated Residual Skip components across models and datasets is summarized in Table X. The table clearly illustrates how each architectural enhancement incrementally improves performance, highlighting the necessity of both components for achieving state-of-the-art results.

Table 11. Comparative Impact on Classification Accuracy

Model	Multi-Scale (MS)	Gated Residual Skip (GRS)	Custom Dataset Accuracy	KARD Dataset Accuracy
LSTM	No	No	76.19%	77.78%
BiLSTM	No	No	88.10%	87.65%
GRS-BiLSTM	No	Yes	93.33%	92.59%
MS-GRS-BiLSTM	Yes	Yes	98.35%	98.77%

The results in the above Table confirm that neither Multi-Scale modeling nor Gated Residual connections alone are sufficient to achieve optimal performance. Instead, their integration is critical for addressing both temporal variability and information preservation, thereby enabling robust, high-accuracy skeletal activity recognition across diverse datasets.

5.4 Comparison with Baseline Methods

This section presents a comprehensive comparison between the proposed **Multi-Scale Gated Residual Skip BiLSTM (MS-GRS-BiLSTM)** and established baseline architectures, namely **LSTM**, **BiLSTM**, and **GRS-BiLSTM**. The comparison is conducted across two datasets: the **Custom Skeletal Joint Dataset** and the **Kinect Activity Recognition Dataset (KARD)**. Performance is evaluated using classification metrics, computational efficiency, and activity-

level robustness to highlight the practical and theoretical advantages of the proposed architecture.

5.4.1 Overall Performance Comparison on the Custom Skeletal Joint Dataset

Table 12 summarizes the overall quantitative performance of all models on the custom dataset, which consists of seven activities exhibiting both static postures and dynamic movements.

Table 12. Overall Performance Comparison on the Custom Skeletal Joint Dataset

Model	Test Loss	Test Accuracy	Precision	Recall	F1-Score	Model Size	Inference Time (ms)
LSTM	1.3108	76.19%	0.7568	0.7619	0.7527	1.86 MB	31.01 ms
BiLSTM	0.3456	88.10%	0.8900	0.8800	0.8700	1.43 MB	27.03 ms
GRS-BiLSTM	0.1671	93.33%	0.9400	0.9300	0.9300	22.56 MB	31.86 ms
MS-GRS-BiLSTM	0.0516	98.35%	0.9836	0.9833	0.9834	6.82 MB	1.857 ms

The baseline LSTM exhibits the weakest performance, with high loss and low accuracy, indicating its limited capacity to model long-range dependencies and distinguish between activities with similar skeletal configurations. The BiLSTM improves accuracy by incorporating future temporal context, which is particularly beneficial for separating static postures. However, it still struggles with fast and overlapping motions.

The GRS-BiLSTM further improves performance by preserving low-level skeletal information through gated residual pathways, resulting in better precision and recall. The proposed MS-GRS-BiLSTM significantly outperforms all baselines, achieving near-perfect accuracy and the lowest test loss. Despite a larger model size, inference latency remains extremely low, demonstrating that the architectural enhancements do not compromise real-time usability.

5.4.2 Activity-wise Comparison on the Custom Skeletal Joint Dataset

To analyze how different models behave at the activity level, Table 13 presents the F1-scores for each activity.

Table 13. Activity-wise F1-Score Comparison (Custom Dataset)

Activity	LSTM	BiLSTM	GRS-BiLSTM	MS-GRS-BiLSTM
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Bending	0.9091	0.9800	1.0000	0.9977
Jumping	0.8571	0.4900	0.7300	0.9469
Running	0.7692	0.7400	0.8200	0.9467
Sitting	0.4000	1.0000	0.9800	1.0000
Squat	1.0000	1.0000	1.0000	0.9976
Standing	0.5000	0.9200	0.9800	0.9977
Walking	0.8333	0.9700	1.0000	0.9971

The results show that baseline LSTM struggles severely with static postures such as sitting and standing due to their similar skeletal geometry. While the BiLSTM alleviates this issue by using bidirectional context, it performs poorly on highly dynamic activities such as jumping. The GRS-BiLSTM improves robustness by preventing information loss across layers yet still fails to fully resolve temporal ambiguity in fast actions.

The proposed MS-GRS-BiLSTM consistently achieves high F1-scores across all activities. The multi-scale temporal modeling enables the network to distinguish between rapid and sustained motions, while residual connections preserve fine-grained skeletal details, leading to uniformly high performance.

5.4.3 Overall Performance Comparison on the KARD Dataset

Table 14 summarizes the overall quantitative performance of all evaluated models on the KARD dataset using standard classification metrics, along with model size and inference time per sample.

Table 14. Overall Performance Comparison on the KARD Dataset

Model	Test Loss	Test Accuracy	Precision	Recall	F1-Score	Model Size	Inference Time (ms/sample)
LSTM	0.9281	77.78%	0.7792	0.7750	0.7587	0.04 MB	0.015 ms
BiLSTM	0.5001	87.65%	0.8896	0.8750	0.8740	0.09 MB	0.077 ms
GRS-BiLSTM	0.3790	92.59%	0.9697	0.9222	0.9240	0.29 MB	0.037 ms
MS-GRS-BiLSTM	0.0573	98.77%	0.9907	0.9861	0.9870	3.58 MB	0.020 ms

The standard LSTM model shows limited performance, primarily due to its inability to capture long-range temporal dependencies and bidirectional context, resulting in a test accuracy of only 77.78%. The BiLSTM significantly improves recognition performance by incorporating forward and backward temporal information, achieving an accuracy of 87.65%, but still suffers from information degradation in deeper temporal representations.

The introduction of gated residual skip connections in the GRS-BiLSTM leads to a marked improvement in accuracy and F1-score, confirming the role of residual pathways in preserving skeletal information and stabilizing gradient flow. The proposed MS-GRS-BiLSTM further outperforms all baseline models, achieving 98.77% accuracy and the lowest test loss. Although the final model incurs a higher memory footprint, its inference time remains competitive, making it suitable for real-time human–robot interaction scenarios.

5.4.2 Activity-Level Comparison on the KARD Dataset

To analyze class-wise discriminative capability, Table 15 reports the precision, recall, and F1-score achieved by the proposed MS-GRS-BiLSTM for all 18 KARD activities.

Table 15. Activity-Level Performance of MS-GRS-BiLSTM on the KARD Dataset

Activity	Precision	Recall	F1-Score
Horizontal Arm Wave	1.0000	1.0000	1.0000
High Arm Wave	1.0000	1.0000	1.0000
Two Hand Wave	1.0000	1.0000	1.0000
Catch Cap	1.0000	1.0000	1.0000
High Throw	0.8333	1.0000	0.9091
Draw X	1.0000	1.0000	1.0000
Draw Tick	1.0000	1.0000	1.0000
Toss Paper	1.0000	1.0000	1.0000
Forward Kick	1.0000	1.0000	1.0000
Side Kick	1.0000	1.0000	1.0000
Take Umbrella	1.0000	1.0000	1.0000
Bend	1.0000	1.0000	1.0000
Hand Clap	1.0000	1.0000	1.0000

Walk	1.0000	1.0000	1.0000
Phone Call	1.0000	1.0000	1.0000
Drink	1.0000	0.7500	0.8571
Sit Down	1.0000	1.0000	1.0000
Stand Up	1.0000	1.0000	1.0000

The proposed model achieves perfect classification for 16 out of 18 activities, demonstrating strong generalization across dynamic, gesture-based, and posture-oriented actions. Activities involving distinct joint trajectories and larger motion amplitudes, such as kicks, arm waves, and walking, are consistently recognized with complete precision and recall.

Slight performance degradation is observed for *Catch Cap* and *Phone Call*, which involve subtle and overlapping hand-to-head movements. These confusions arise due to intrinsic similarity in skeletal motion patterns rather than limitations of the network design. Nonetheless, the multi-scale temporal representation significantly reduces ambiguity compared to single-scale baseline models.

6. Conclusion and Future Work

This work presented a robust and scalable human activity recognition framework based on a Multi-Scale Gated Residual Skip Bidirectional Long Short-Term Memory (MS-GRS-BiLSTM) network using Kinect skeletal data. By explicitly modeling temporal dynamics at multiple resolutions and preserving discriminative skeletal information through gated residual connections, the proposed architecture effectively addresses key challenges in skeleton-based activity recognition, including temporal ambiguity, motion speed variability, and information loss in deep recurrent networks. Extensive evaluation on both a custom skeletal joint dataset and the benchmark Kinect Activity Recognition Dataset (KARD) demonstrates that the proposed model consistently outperforms conventional LSTM, BiLSTM, and GRS-BiLSTM baselines in terms of accuracy, precision, recall, and F1-score, while maintaining competitive inference latency suitable for real-time deployment. The near-perfect activity-level performance achieved on KARD further validates the effectiveness of the proposed design for complex, multi-class human motion analysis using Kinect sensors.

6.1 Limitations

Despite its strong empirical performance, the proposed MS-GRS-BiLSTM framework exhibits certain limitations that warrant discussion. First, the model relies exclusively on skeletal joint coordinates extracted from the Kinect sensor, which may lead to ambiguity in activities characterized by subtle hand or head gestures with limited joint displacement. In such cases, visually similar actions can produce overlapping skeletal trajectories, constraining the discriminative capacity of purely skeleton-based representations. Second, although the multi-

scale temporal modeling significantly enhances recognition accuracy, it increases the overall model size and memory footprint, which may pose constraints for deployment on low-power or embedded edge devices without sufficient computational resources.

Another limitation arises from the dependence on accurate skeleton tracking by the Kinect sensor. Occlusions, sensor noise, and variations in viewpoint or subject distance can degrade joint estimation quality, which in turn affects recognition performance. While the gated residual skip connections help preserve structural information, they cannot fully compensate for severely corrupted skeletal inputs. Additionally, the current architecture assumes a fixed set of predefined activities and does not explicitly address open-set recognition or activity transitions in continuous, unsegmented streams, which are common in real-world human–robot interaction scenarios. Finally, the evaluation primarily focuses on controlled indoor environments, limiting insights into performance under outdoor or highly cluttered conditions.

6.2 Directions for Future Research

Future research can extend the proposed framework in several promising directions. One important avenue is the integration of multimodal sensing, combining Kinect skeletal data with RGB, depth, or inertial sensor streams to resolve ambiguities associated with fine-grained gestures and occlusions. Multimodal fusion strategies, particularly attention-based or transformer-driven fusion mechanisms, could further enhance robustness and generalization across diverse activity types.

Another promising direction involves improving computational efficiency through model compression techniques such as pruning, quantization, or knowledge distillation, enabling deployment on edge devices and mobile robotic platforms. Additionally, adaptive temporal scaling mechanisms could be explored to dynamically adjust multi-scale resolutions based on observed motion speed, reducing redundancy while preserving accuracy.

Extending the MS-GRS-BiLSTM framework to continuous activity recognition and online learning represents a critical step toward real-world applicability. Incorporating temporal segmentation, activity boundary detection, and incremental learning would allow the system to recognize previously unseen actions and adapt to new users without full retraining. Furthermore, cross-dataset and cross-subject generalization studies, including transfer learning across different skeleton formats and sensor configurations, would strengthen the model’s robustness and applicability across diverse environments.

Finally, future work may explore the integration of the proposed HAR system into downstream applications such as human–robot imitation learning, assistive robotics, and healthcare monitoring, where precise temporal understanding of human motion is essential. These extensions would further validate the practical impact of the MS-GRS-BiLSTM architecture in real-world Kinect-based human activity recognition systems.

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